

Homework-1: System Setup, GitHub Skill Review, and Introduction to Agentic Engineering

Please upload your PDF file to Gradescope by **September 10, 11:59PM ET**.

This homework is about getting your development environment ready for future assignments. You'll install and configure the tools we use all semester. Along the way you'll pick up the basic ideas behind agentic engineering and get some practice with GitHub, which is used submit future assignments.

Upload your submission Gradescope as one PDF containing every screenshot and detail requested below. Map your submission pages to the corresponding parts on Gradescope.

Permitted AI Usage: You may use AI tools to debug any installation problems. If you do, include the transcript of your AI conversation (or a series of screenshots showing your prompts to the AI tool and its responses) at the end of your submission. **Do NOT have an AI tool or agent perform the installation for you.** You may type in commands that an AI tool suggested, but you are not allowed to let an AI agent run them on your machine.

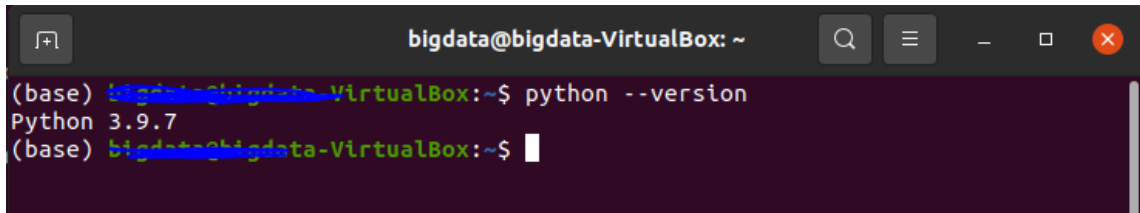
Part-I (60%): Setup Course Tech Stack Environment

In this part, you are expected to install several applications and make sure they run properly. You are asked to submit a screenshot for every application to prove that it's working properly.

- It is highly recommended that you review the installation tutorials & video guides published on canvas and attend the system installation office hours corresponding to your operating system. The materials will be made available starting 8/27 and environment setup office hours are scheduled on Friday 8/28 at the following times (over Zoom using TA OHs URL):
 - Mac: 2-3pm
 - Linux: 4-5pm
 - Windows: 6-7pm
- Each screenshot must clearly demonstrate that it was taken on your own machine. For example, you may include your username, computer information, or your name typed in a Notepad file visible in the screenshot.
- Don't use cloud-hosted [virtual machines](#) to install or run the required software.
- Download the latest available software version that is compatible with your operating system.
- Sample screenshots are provided in the following pages.

You need to install the following applications:

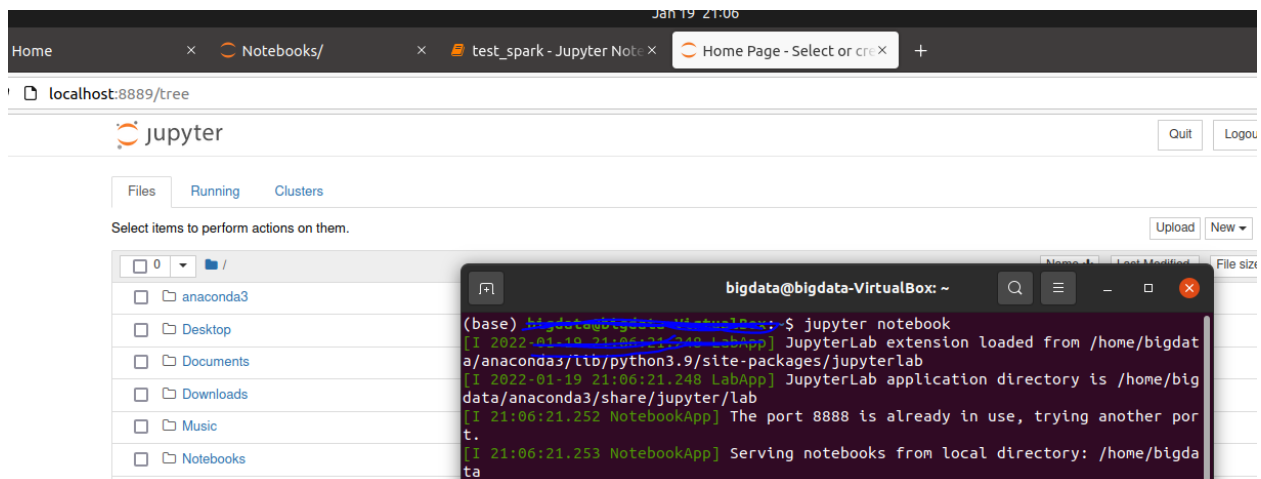
1. [Anaconda Python](#) 3.x. Submit a screenshot showing the version (username was omitted to avoid reusability).



```
bigdata@bigdata-VirtualBox: ~  
(base) bigdata@bigdata-VirtualBox:~$ python --version  
Python 3.9.7  
(base) bigdata@bigdata-VirtualBox:~$
```

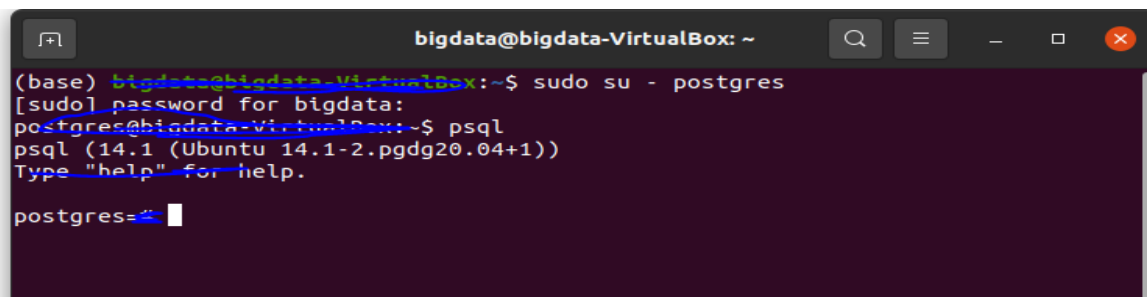
2. [Jupyter](#). Make sure you can open and run Jupyter notebooks.

Note: In my screenshot, I launched Jupyter Notebook from the terminal. Your approach might be different. Be sure to clearly show and explain how you are launching Jupyter Notebook in your own screenshot. You are welcome to use more than one screenshot if needed.



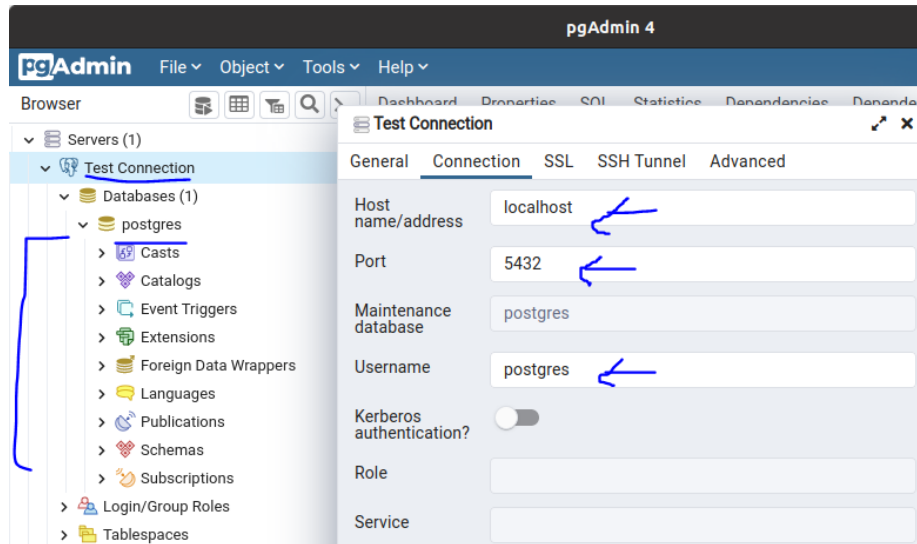
3. [PostgreSQL](#). Submit a screenshot showing how you connect to the database via the terminal or shell.

Note: The command you use may differ depending on your operating system. On Windows, you can use SQL Shell (psql) to connect to PostgreSQL



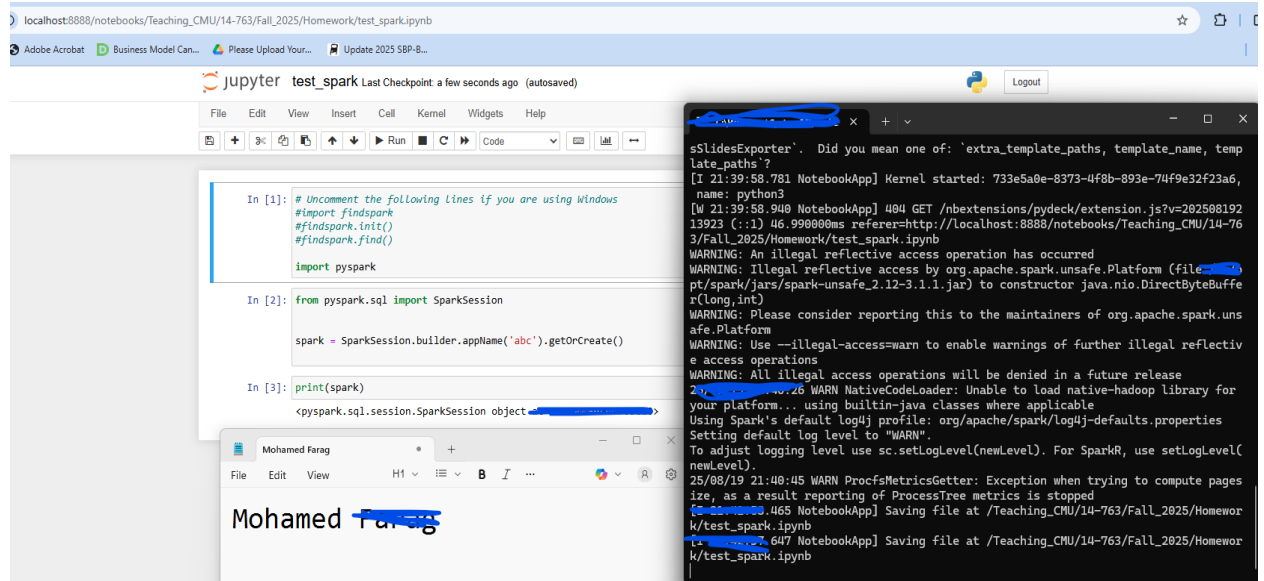
```
bigdata@bigdata-VirtualBox: ~  
(base) bigdata@bigdata-VirtualBox:~$ sudo su - postgres  
[sudo] password for bigdata:  
postgres@bigdata-VirtualBox:~$ psql  
psql (14.1 (Ubuntu 14.1-2.pgdg20.04+1))  
Type "help" for help.  
  
postgres=#
```

4. **PgAdmin4.** PgAdmin4 is a Graphical User Interface for PostgreSQL. Post a screenshot showing a successful connection from PgAdmin4 to PostgreSQL.



5. **Apache Spark.** We will use Apache Spark for big data processing and machine learning. Run the test_spark notebook file (posted on the assignment page in Canvas) and submit a screenshot of its successful execution.

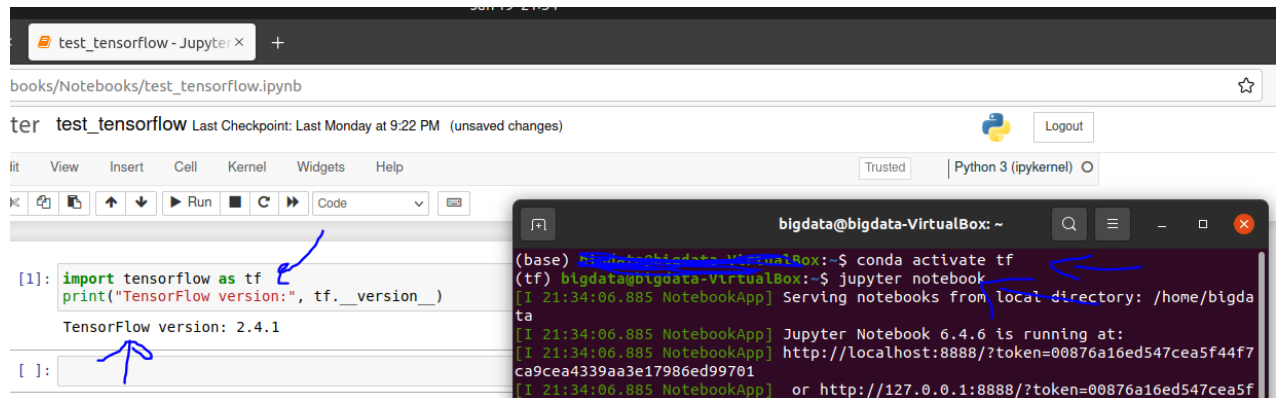
Note: In my screenshot, I launched Spark from the terminal using the command **jupyter notebook**. Your approach may be different. Please make sure to clearly show how you are launching Spark and display the output of the notebook in your screenshot(s). You may include more than one screenshot if needed. Use the latest available Spark version for your operating system.



6. [TensorFlow](#). We will use TensorFlow for machine learning modeling. Run the test_tensorflow notebook (posted on the assignment page in Canvas) and submit a screenshot showing its successful execution.

Hint:

1. If you can't run TensorFlow in your default python environment, you may consider creating a virtual environment.



The screenshot shows a Jupyter Notebook interface with a code cell containing the following code:

```
[1]: import tensorflow as tf
    print("TensorFlow version:", tf.__version__)
```

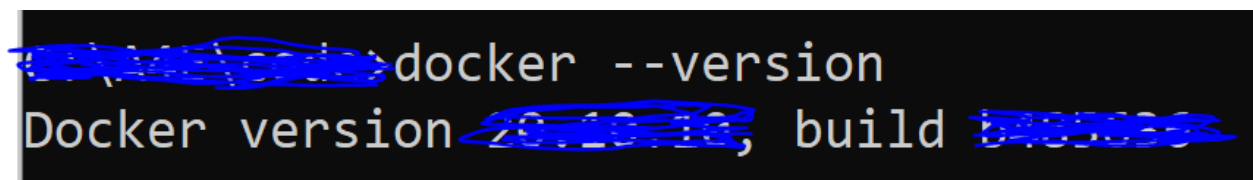
The output of the code cell is:

```
TensorFlow version: 2.4.1
```

Below the code cell, there is a terminal window showing the following commands and output:

```
(base) bigdata@bigdata-VirtualBox:~$ conda activate tf
(tf) bigdata@bigdata-VirtualBox:~$ jupyter notebook
[I 21:34:06.885 NotebookApp] Serving notebooks from local directory: /home/bigdata
[I 21:34:06.885 NotebookApp] Jupyter Notebook 6.4.6 is running at:
[I 21:34:06.885 NotebookApp] http://localhost:8888/?token=00876a16ed547cea5f44f7ca9cea4339aa3e17986ed99701
[I 21:34:06.885 NotebookApp] or http://127.0.0.1:8888/?token=00876a16ed547cea5f44f7ca9cea4339aa3e17986ed99701
```

7. [Pytorch](#). We will use PyTorch for building neural networks. In your Jupyter notebook, import PyTorch without errors. Ensure that both the **pyspark** and **torch** packages can be imported within the same notebook. Submit a screenshot showing the successful execution of these imports.
8. [Docker](#). We will use Docker for NoSQL databases. Submit a screenshot that reflects the version of your docker.



The screenshot shows a terminal window with the following command and output:

```
bigdata@bigdata-VirtualBox:~$ docker --version
Docker version 20.10.10, build b3b0673
```

Part-II (10%): Get familiar with Git and GitHub

In this section, you will refresh your memory about Git and GitHub.

- If you haven't used version control systems before, you can watch the following videos:
 - Version Control Systems Overview: <https://www.youtube.com/embed/ils1aehi3VU>
 - Git and GitHub: <https://www.youtube.com/embed/91ZZv002etU>

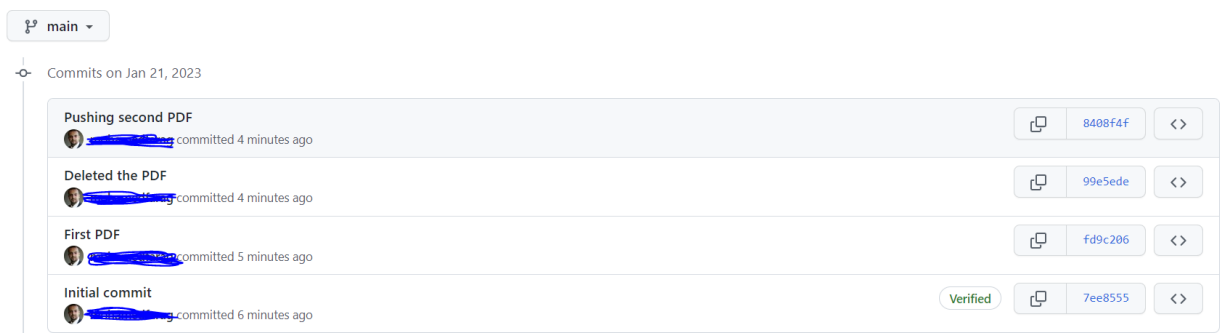
I. Submit screenshots for the completion of all the following GitHub exercises:

- a) <https://github.com/skills/introduction-to-github>
- b) <https://github.com/skills/communicate-using-markdown>

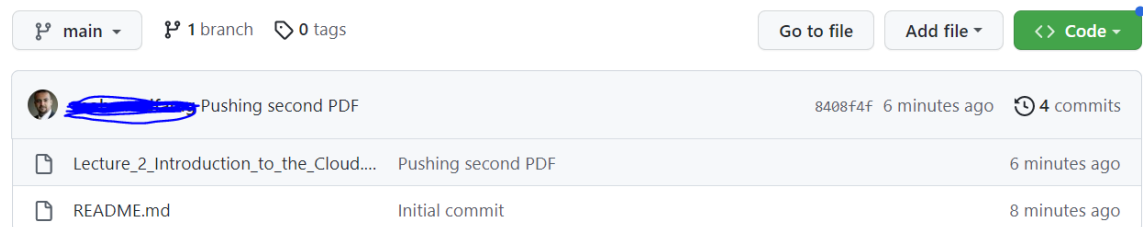
Your screenshots should reflect successful completion of the above labs.

II. Also, complete the following task:

- Create a public GitHub repository named "Test-Toolchains".
- Include a README file when creating the repository.
- Commit and push the PDF of the first lecture to the repository.
- In a separate commit, delete the first lecture PDF and push the change.
- Commit and push a different PDF file to the repository. Submit the following:
 - a) A screenshot of your GitHub repository history



b) A screenshot of the current view of your repository



c) The URL of your **public** repository

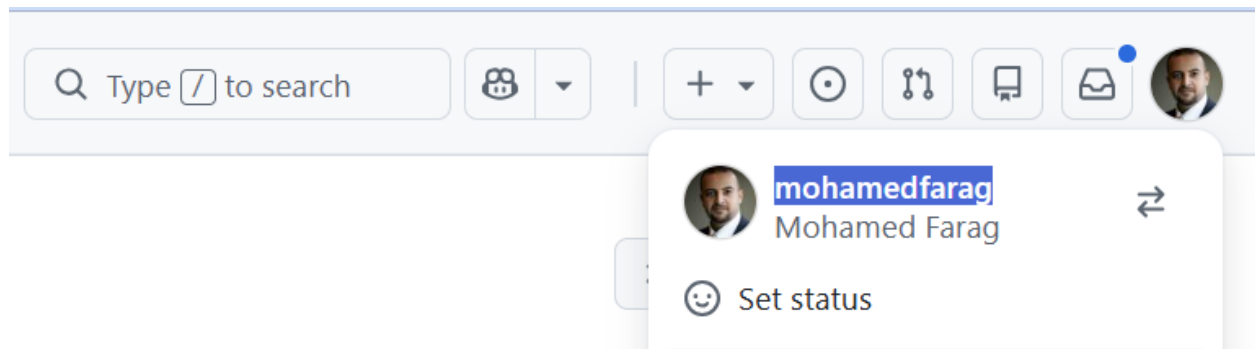
Part-III (10%): Submit your GitHub username.

In this section, you will submit your GitHub account username via this Google form:

<https://forms.gle/XBjENsxNfQu78hGz8>

To access the form, you must be logged in with your Andrew email. We will use your GitHub username to invite you to classroom50, a tool that is used to manage student GitHub repositories for future assignments.

You can find your GitHub account username by clicking your account photo/logo in the top right corner of the screen (as shown in the screenshot below)



Please don't provide your email address, name, or any other personal information. In the example above, the correct answer to submit is: mohamedfarag

Don't provide any prefixes or suffixes in your answer. Only submit your username (without periods, quotes, etc.)

Part-IV (20%): Get familiar with the pros and cons of vibe coding.

In this section, you will learn some basics about agentic engineering.

Watch the following videos and answer the questions below:

- Video 1: <https://www.youtube.com/watch?v=fjZdqNJ7UzY>
- Video 2: <https://www.youtube.com/watch?v=eCqHGMFjD9U>
- Video 3: <https://www.youtube.com/watch?v=9GUw-QBenCU>

In your own words, answer the following questions:

1. Explain the risks associated with vibe coding.
2. What safe use cases for vibe coding were discussed in the assigned videos?
3. Explain the agent loop used by coding agents.
 - From the assigned videos, cite the diagram which illustrates the agent loop.
4. What is the difference between prompt injection and jailbreaking? Explain the distinction using the example provided in the assigned videos.

Note: The videos provided are extracted from the AGAI group mini-capsule series. The full series videos can be accessed via this URL: <https://www.cmu.edu/ini/agai/summer-2026-mini-capsules/>